

Pharmaniaga Aciclovir Tablet

DESCRIPTION

Pharmaniaga Aciclovir Tablet 200 mg

Yellow, shield-shaped, bevelled edge tablet with Pharmaniaga icon on one side and plain on the other.

Pharmaniaga Aciclovir Tablet 400 mg

Pink, oblong shaped tablet with Pharmaniaga icon on one side and plain on the other.

COMPOSITION

Each tablet contains Aciclovir 200 mg or 400 mg.

ACTIONS

Aciclovir is converted to aciclovir monophosphate, a nucleotide, by the viral thymidine kinase of herpes simplex virus (HSV) and varicella-zoster virus (VZV). Aciclovir monophosphate is converted to the diphosphate by cellular guanylate kinase and to the triphosphate by a number of cellular enzymes. Aciclovir triphosphate interferes with HSV and VZV DNA polymerase and inhibits viral DNA replication. The triphosphate can be incorporated into growing chains of DNA by viral DNA polymerase, resulting in termination of the DNA chain.

Absorption: Bioavailability 20 % (range, 15 to 30 %). Poorly absorbed from the gastrointestinal tract. Not significantly affected by food.

Distribution: Widely distributed to tissues and body fluids, including brain, kidneys, lungs, liver, aqueous humor, tears, intestines, muscle, spleen, breast milk, uterus, vaginal mucosa, vaginal secretions, semen, amniotic fluid, cerebrospinal fluid (CSF) and herpetic vesicular fluid. Highest concentrations are found in kidneys, liver, and intestines. CSF concentrations are approximately 50% of plasma concentrations. Crosses the placenta, also.

Adults: Approximately 48 liters (L) per square meter of body surface (m²) (range, 37 to 57 L per m²)

Children and adolescents (1 to 18 years old): Approximately 45 L per m².

Neonates (0 to 3 month old): Approximately 28 L per m² (range, 24 to 30 L per m²).

End stage renal disease: Approximately 41 L per m².

Protein binding: Low (9 to 33 %)

Biotransformation: Hepatic, only major metabolites found in urine is 9-carboxymethoxymethylguanine, which accounts for approximately 9 to 14 % of the dose. This metabolite has no known antiviral activity.

Half-life: Oral- 3.3 hours

Time to peak serum concentration: Oral - 1.7 hours.

Renal impairment (adults)

Creatinine Clearance (mL/min)/(mL/sec)	Half-life (hr)
> 80/1.33	2.5
50-80/0.83 — 1.33	3.0
15-50/0.25-0.83	3.5
Anuric	19.5
During haemodialysis	5.7
Continuous ambulatory peritoneal dialysis	14-18

Elimination:

- Renal : Excreted by both glomerular filtration and tubular secretion.
- Oral : Approximately 14% of total dose excreted unchanged in urine.
- Faecal : Insignificant amounts (< 2%).
- Lungs : Trace amounts in exhaled CO₂.
- Dialysis : Haemodialysis : A single 6-hour period of haemodialysis reduces plasma acyclovir concentration by approximately 60 %.
- Peritoneal : Peritoneal dialysis does not substantially alter aciclovir clearance.

INDICATIONS

Treatment and prophylaxis of herpes genitalis; treatment of herpes zoster, herpes simplex and varicella (chicken-pox)

CONTRAINDICATIONS

Aciclovir is contraindicated in patients who have developed hypersensitivity or intolerance to the drug.

ADVERSE REACTIONS

General: fever, headache, pain, peripheral oedema, and rarely anaphylaxis.

Nervous: confusion, dizziness, hallucinations, paraesthesia, somnolence (These symptoms may be marked, particularly in older adults).

Digestive: diarrhoea, elevated liver function test, gastro-intestinal distress, nausea.

Haemic and Lymphatic: leucopenia, lymphadenopathy

Musculoskeletal: myalgia.

Skin: alopecia, pruritus, rash, urticaria.

Special Senses: visual abnormalities.

Urogenital: elevated creatinine

PRECAUTIONS

Acute renal insufficiency may occur due to precipitation of aciclovir in the renal tubules. It is most likely to occur if aciclovir is given by rapid intravenous injection, concurrently with known nephrotoxic medications, to patients who are inadequately hydrated, or to patients with renal function impairment without appropriate dosage reduction. However, acute renal failure has also been reported in patients receiving oral aciclovir.

Neuropsychiatric toxicity has been associated with high plasma aciclovir concentrations, which may occur when high doses are used, or when patients with renal function impairment are not given an appropriately lowered dose. Neuropsychiatric toxicity may also be more likely to occur in immunocompromised patients and geriatric patients.

Pregnancy / Reproduction

Fertility - Impairment of spermatogenesis, sperm motility, or morphology has not been documented in humans. However, high doses of parenteral aciclovir have caused testicular atrophy in rats and dogs. Some evidence of sperm production recovery was evident 30 days post-dose.

Pregnancy - Aciclovir crosses the placenta. Aciclovir has been used in all stages of pregnancy, most commonly in the third trimester. No adverse foetal effects have been reported.

Aciclovir should not be used during pregnancy unless the potential benefit justifies the potential risk to the fetus.

Breast-feeding

Aciclovir passes into breast milk at concentrations from 0.6 to 4.2 times the corresponding plasma concentration. A very small amount of aciclovir has

been measured in one nursing infant's urine; no toxicity was observed. Caution should be exercised when aciclovir is administered to a nursing mother.

Paediatrics

Limited data is available about the use of oral aciclovir in children younger than 2 years of age.

Geriatrics

Elderly patients are more likely to have an age-related decrease in renal function, which may require an adjustment of aciclovir dosage or dosing interval.

Laboratory value

Blood urea nitrogen (BUN) and serum creatinine concentrations may be increased.

DRUG INTERACTIONS

Nephrotoxic medications: concurrent use may increase the potential for nephrotoxicity, especially in the presence of renal function impairment.

Probenecid: concurrent use may decrease renal tubular secretion of aciclovir resulting in increased aciclovir serum and CSF concentrations, prolonged elimination half-life and potentially, increased toxicity.

Effects on Ability to Drive and Use Machine
There have been no studies to investigate the effect of aciclovir on driving performance or the ability to operate machinery.

DOSAGE AND ADMINISTRATION

Adults:

Genital herpes infections-

Initial therapy: Oral, 200 mg every four hours while awake, five times a day, for ten days.

Recurrent infections, intermittent therapy: Oral, 200 mg every four hours while awake, five times a day, for five days.

Recurrent infections, chronic suppressive therapy: Oral, 400 mg twice a day; or 200 mg three to five times a day.

Herpes zoster-

Oral, 800 mg five times a day, for seven days.

Varicella (Chicken-pox)-

Oral, 20 mg per kg of body-weight, up to 800 mg per dose, five times a day for seven days. Treatment

should be initiated at the earliest sign or symptom of chicken-pox.

The recommended dose for initial therapy and intermittent therapy of herpes infections patients with renal function impairment is:

Creatinine Clearance (mL/min)/(mL/sec)	Dose (mg)	Dosing Interval (hr)
Genital herpes : Initial/intermittent therapy >10/0.17	200	4 (5 times daily)
0-10/0-0.17	200	12
Chronic suppressive >10/0.17	400	12
0-10/0-0.17	200	12
Herpes zoster (shingles) : >25/0.42	800	4 (5 times daily)
10-25/0.17-0.42	800	8
0-10/0-0.17	800	12

Children:

Varicella (Chicken-pox) -

20 mg per kg (maximum 800 mg) 4 times daily,
or

or

Under 2 years : 200 mg 4 times daily.

2 - 5 years : 400 mg 4 times daily.

Over 6 years : 800 mg 4 times daily.

OVERDOSAGE

Symptoms : Patients have ingested up to 20 g of acyclovir with no unexpected adverse effects. Anuria and acute renal failure may occur.

Treatment : Since there is no specific antidote; treatment of adverse effects should be symptomatic and supportive. Adequate hydration may be required to prevent precipitation of acyclovir in the renal tubules. Haemodialysis may aid in the removal of aciclovir from the blood, especially in patients with acute renal failure and anuria.

ROUTE OF ADMINISTRATION

Oral.

STORAGE CONDITIONS

Store below 30°C.

Protect from light.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

Pharmaniaga Aciclovir Tablet 200 mg

In boxes of 25 tablets (5x5's) and 50 tablets (5x10's).

Pharmaniaga Aciclovir Tablet 400 mg

In boxes of 50 tablets (5x10's).

Date of revision: 06th October 2022

PRODUCT	REGISTRATION	HOLDER/
MANUFACTURER:		
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